1. Suppose there are two production lines *A* and *B* for producing bagged food, where the standard weight for one bagged food is 50g. Due to many factors, the bagged food produced by these two production lines may suffer from weight errors, both of which follow Gaussian distributions with $error_A \sim N(1,4)$ and $error_B \sim N(0,6.25)$. If we have equal prior probability on either production line, given a bagged food with weight 52g, which production line does it belong to?

Solution:

We have $x = 52 - 50 = 2$

$$P(x|A) = \frac{1}{\sqrt{2\pi} \cdot 2} e^{-\frac{(x-1)^2}{2 \cdot 2^2}} \approx 0.1760$$

$$P(x|B) = \frac{1}{\sqrt{2\pi} \cdot 2.5} e^{-\frac{(x-0)^2}{2 \cdot 2.5^2}} \approx 0.1159$$

According to,

$$\mathrm{P}(A|x) = \frac{P(x|A)P(A)}{P(x|A)P(A) + P(x|B)P(B)}$$

$$\mathrm{P}(B|x) = \frac{P(x|B)P(B)}{P(x|A)P(A) + P(x|B)P(B)}$$

And we have $P(A) = P(B)$,
As $P(x|A) > P(x|B)$, so $\mathrm{P}(A|x) > \mathrm{P}(B|x)$
Therefore, it is more likely to belong to the production line A.

2. There are three chicken *A*, *B*, and *C* caught by a fox. One of them will be killed by the fox in the next morning when the sun rises. Chicken *A* feels very nervous as he has 1/3 chance to be the one being killed tomorrow. He tries to get some information from the fox: "I know you cannot tell me whether I will be killed in the next morning. But can you tell me which of my mates *B* and *C* will not be killed?" Accordingly, the fox who tells *A* that *C* won't be killed. At a second thought, *A* gets much more worried. Before he asked the fox, he thought he had 1/3 chance being killed, but with *C* excluded, the chance of himself being killed increases to 1/2.

Let E_X $(X \in \{A, B, C\})$ denote the event that X will be killed tomorrow. By assuming equal prior probability, we have $P(E_A) = P(E_B) = P(E_C) = 1/3$.

Given the above problem descriptions, please try to answer the following questions:

(a) Let L_Y $(Y \in \{B, C\})$ denote the event that the fox tells A that Y will not be killed tomorrow (the fox won't tell A's fate, so $A \notin Y$), and E_Z $(Z \in \{A, B.C\})$ denote the event that Z will be killed tomorrow. What are the values for the conditional probability $P(L_B|E_A)$, $P(L_C|E_A)$, $P(L_B|E_B)$, $P(L_C|E_B)$, $P(L_B|E_C)$ and $P(L_C|E_C)$?

(b) Calculate the posterior probability of A being killed given that the fox tells A that C will not be killed tomorrow, i.e. $P(E_A|L_C)$. Did the fox tell us any more information about A's fate?

(1)

Let L_Y, where Y = {B, C}, denote the event that: knowing that X will be executed tomorrow, the likelihood of the janitor telling A that Y will live. The likelihoods are: $P(L_B|E_A) = P(L_C|E_A) = \frac{1}{2}$; $P(L_B|E_B) = 0, P(L_C|E_B) = 1$; $P(L_B|E_C) = 1, P(L_C|E_C) = 0$.

(2)

Solution:
The posterior probability of A being executed is

$$\begin{aligned} P(E_A|L_C) &= \frac{P(L_C|E_A) \cdot P(E_A)}{P(L_C)} \\ &= \frac{P(L_C|E_A) \cdot P(E_A)}{P(L_C|E_A) \cdot P(E_A) + P(L_C|E_B) \cdot P(E_B) + P(L_C|E_C) \cdot P(E_C)} \\ &= \frac{1/2 \cdot 1/3}{1/2 \cdot 1/3 + 1 \cdot 1/3 + 0 \cdot 1/3} \\ &= \frac{1}{3}. \end{aligned}$$

NO

3. Let $\{x_i \mid 1 \le i \le n\}$ be the set of *i.i.d.* examples randomly drawn from a Bernoulli distribution with parameter p $(0 \le p \le 1)$, i.e. $P(x|p) = p^x(1-p)^{1-x}$ $(x \in \{0,1\})$. What is the maximum likelihood estimation $\hat{p}$ for the parameter given $\{x_i \mid 1 \le i \le n\}$?
3.

$$f(x|p) = p^x(1-p)^{1-x}, x \in \{0,1\}$$

$$L(p) = \prod_{i=1}^{n} f(x_i|p) = p^{\Sigma_{i=1}^n x_i}(1-p)^{n-\Sigma_{i=1}^n x_i}$$

$$\ln L(p) = \left(\sum_{i=1}^{n} x_i\right) \cdot \ln p + \left(n - \sum_{i=1}^{n} x_i\right) \ln(1-p)$$

$$\frac{d \ln L(p)}{dp} = \frac{\sum_{i=1}^{n} x_i}{p} - \frac{n - \sum_{i=1}^{n} x_i}{1-p} = 0$$

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

4. Given a hidden Markov model $\boldsymbol{\theta} = (\mathbf{A}, \mathbf{B}, \boldsymbol{\pi})$ with three hidden states $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and two visible symbols $\mathcal{V} = \{v_1, v_2\}$. Specifically, the HMM parameters correspond to:

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.1 & 0.4 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \\ 0.7 & 0.3 \end{bmatrix}, \qquad \boldsymbol{\pi} = (0.2, 0.3, 0.5)$$

Based on the above problem setting, please try to answer the following questions:

(a) Given two symbol sequences $\mathbf{V_1} = \{v_1, v_2, v_1\}$ and $\mathbf{V}_2 = \{v_1, v_2, v_2\}$, which sequence is more likely to be observed?

(b) Based on the Viterbi algorithm, determine the most likely state sequence $\boldsymbol{\omega}$ which leads to the observed symbol sequence $\mathbf{V} = \{v_1, v_2, v_1\}$.

(1)

Use forward algorithm to calculate $\mathrm{P}(\boldsymbol{X}_1|\lambda)$:

$\alpha_1(1) = \pi_1 b_{1x(1)} = 0.2 * 0.5 = 0.1,$

$\alpha_2(1) = 0.3 * 0.4 = 0.12, \alpha_3(1) = 0.5 * 0.7 = 0.35$

$$\alpha_i(t+1) = \left[\sum_{j=1}^{3} \alpha_j(t) a_{ji}\right] b_{ix(t+1)}$$

	t=1	t=2	t=3
i=1	0.1	0.078	0.040
i=2	0.12	0.084	0.026
i=3	0.35	0.082	0.068

$$\mathrm{P}(X_1|\lambda) = \sum_{i=1}^{3} \alpha_i(3) = 0.134$$

In the same way, $\mathrm{P}(\boldsymbol{X}_2|\lambda) = 0.109$

So, the sequence $\mathbf{X_1}$ is more likely to be observed.

(2)

Use Viterbi algorithm to find the most likely sequence of hidden states:

$\delta_i(1) = \pi_i b_{ix(1)}$

$\delta_1(1) = 0.1, \delta_2(1) = 0.12, \delta_3(1) = 0.35$

$$\delta_i(2) = \max_{1 \le j \le 3} \left[\delta_j(1) a_{ji}\right] b_{ix(2)}$$

$$\delta_1(2) = 0.035, \delta_2(2) = 0.036, \delta_3(2) = 0.063$$

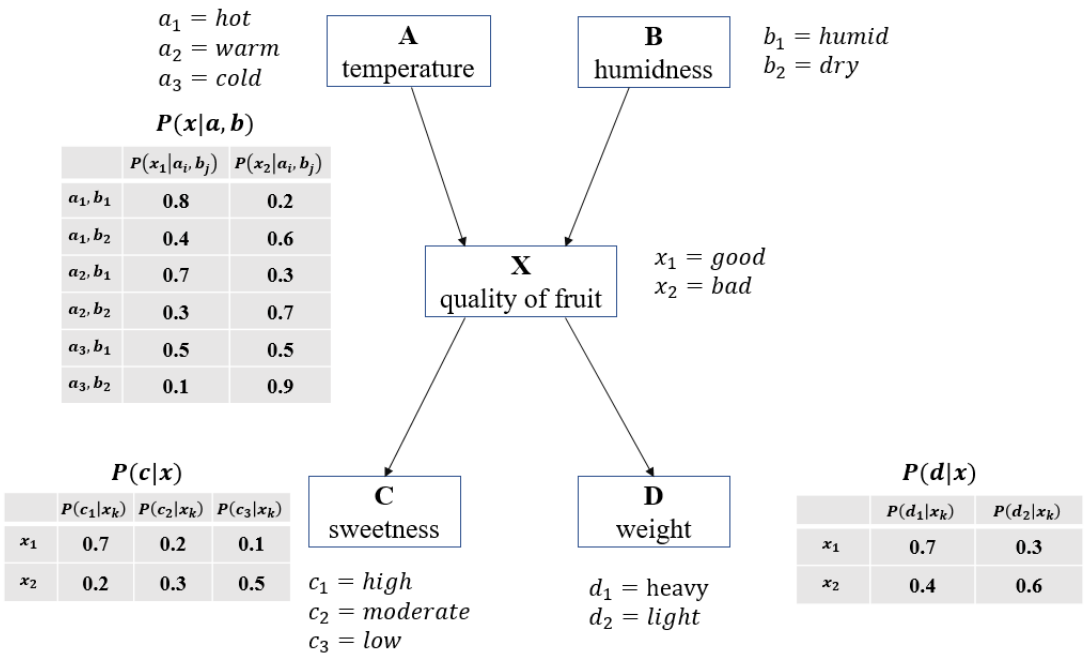
$$\psi_i(2) = arg \max_{1 \le j \le 3} \delta_j(1) a_{ji}$$

$$\psi_1(2) = 3, \psi_2(2) = 3, \psi_3(2) = 3$$

$$\delta_1(3) = 0.0088, \delta_2(3) = 0.0072, \delta_3(3) = 0.0265$$
$$\psi_1(3) = 1, \psi_2(3) = 2, \psi_3(3) = 3$$

So, the most likely sequence of hidden states is $\boldsymbol{\omega} = \{\omega_3, \omega_3, \omega_3\}$

5. Temperature and humidness influence the quality of the fruit, while sweetness and weight are two important factors to evaluate the quality of the fruit. Suppose we have a Bayesian belief network shown as below to model the above phenomenon:



$\boldsymbol{P(x|a,b)}$

	$P(x_1\|a_i,b_j)$	$P(x_2\|a_i,b_j)$
a_1,b_1	0.8	0.2
a_1,b_2	0.4	0.6
a_2,b_1	0.7	0.3
a_2,b_2	0.3	0.7
a_3,b_1	0.5	0.5
a_3,b_2	0.1	0.9

$\boldsymbol{P(c|x)}$

	$P(c_1\|x_k)$	$P(c_2\|x_k)$	$P(c_3\|x_k)$
x_1	0.7	0.2	0.1
x_2	0.2	0.3	0.5

$\boldsymbol{P(d|x)}$

	$P(d_1\|x_k)$	$P(d_2\|x_k)$
x_1	0.7	0.3
x_2	0.4	0.6

Accordingly, please try to answer the following questions:

(a) Suppose we have $P(a_1)$=0.8, $P(a_2)$=0.2, $P(a_3)$=0, and $P(b_1)$=1.0, $P(b_2)$=0. Given that the fruit grows up in *hot* and *humid* environment and has *high* sweetness, what would the quality of the fruit be?

(b) Suppose we have $P(b_1)$=0.5, $P(b_2)$=0.5. Given a box of fruit grown up in *hot* environment, what is the probability of finding a fruit with *moderate* sweetness and *heavy* weight in this

5(1) (这一小问答案与题目不符)
结果为 0.448\alpha 和 0.032\alpha （未 Normalize 之前）

$$\mathrm{P}(x_1|\boldsymbol{e}) = \alpha \sum_{a,d} P(x_1, b_1, c_1, a, d)$$

$$= \alpha \mathrm{P}(b_1) P(c_1|x_1) \sum_a P(a) P(x_1|a, b_1) \sum_d P(d|x_1) = 0.5460\alpha$$

$$\mathrm{P}(x_2|\boldsymbol{e}) = \alpha \sum_{a,d} P(x_2, b_1, c_1, a, d)$$

$$= \alpha \mathrm{P}(b_1) P(c_1|x_2) \sum_a P(a) P(x_2|a, b_1) \sum_d P(d|x_2) = 0.044\alpha$$

Normalize, $\mathrm{P}(x_1|\boldsymbol{e}) = 0.9254, \mathrm{P}(x_2|\boldsymbol{e}) = 0.0746$
$\mathrm{P}(x_1|\boldsymbol{e}) > \mathrm{P}(x_2|\boldsymbol{e})$, so the fruit is more likely to have good quality.
(2)

$$\mathrm{P}(c_2, d_1|a_1) = \frac{\mathrm{P}(a_1, c_2, d_1)}{P(a_1)} = \frac{\sum_{b,x} \mathrm{P}(a_1, c_2, d_1, b, x)}{P(a_1)}$$

$$= \frac{P(a_1) \sum_{b,x} \mathrm{P}(b) P(x|a_1, b) P(c_2|x) P(d_1|x)}{P(a_1)}$$

$$= \sum_{b,x} \mathrm{P}(b) P(x|a_1, b) P(c_2|x) P(d_1|x) = 0.132$$

Answers:

1.

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